When the protein differs from the code

Alternate RNA decoding results in stable and abundant proteins in mammals

Tsour et al. — *Nature* 656, 506–515 (2026)

Journal Club | doi:10.1038/s41586-026-10678-2

The question: can rare decoding events make abundant proteins?

- A protein can differ from its reference sequence through DNA variation, RNA editing, or incorporation of a different amino acid during translation.
- The last mechanism is usually treated as rare error. But the steady-state abundance of its product depends on *both* synthesis and removal.
- This paper asks how often such products are detectable in mammalian tissues, how abundant they become, and what explains that abundance.

Central result
Some substituted products are more abundant than the corresponding encoded product. Their stability contributes to this accumulation.

Introduction; Figs. 1–3. "Alternate translation" is the authors' neutral term; adaptive function is not assumed.

What is actually measured?

BP: base peptide with the encoded amino acid.

SAAP: substituted amino acid peptide.

Shared peptides: other peptides common to the two inferred proteoforms.

$$\mathrm{RAAS} = \frac{\text{SAAP abundance}}{\text{BP abundance}}$$

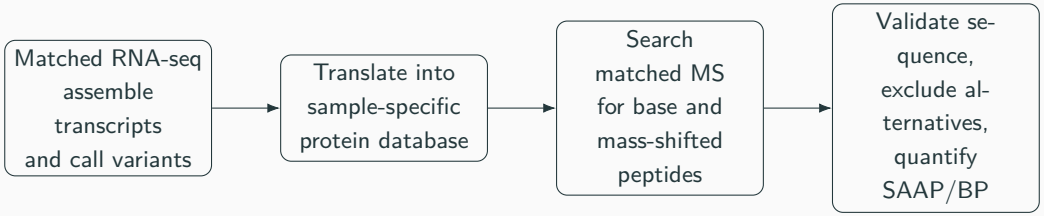
A toy measurement
BP intensity $= 100$
SAAP intensity $= 600$
RAAS $= 6$

If peptide response factors are comparable, the substituted product dominates at this site.

A peptide ratio estimates a proteoform ratio; it does not sequence a complete intact protein or directly measure decoding probability.

Fig. 2a; Methods: Quantifying SAAPs, BPs and RAAS. Numerical example is illustrative.

A search designed to separate translation from sequence variation



- 1,094 human samples: six cancer types with adjacent normal tissue, plus 26 healthy tissue types.
- TMT and label-free MS from independent studies; paired DNA analysis removes mutation-compatible candidates.
- Discovery uses systematic precursor and fragment mass shifts relative to an identified base peptide (MaxQuant dependent-peptide search).

Fig. 1a,b; Methods: Sample-specific protein databases, Proteomic data processing, Genomic mutation analysis.

The difficult step is excluding alternative explanations

1. Reject mass shifts compatible with known PTMs; e.g., N→D and Q→E cannot be separated from deamidation here.
2. Re-search the candidate sequences in a standard database search; require at least two substitution-supporting fragment ions.
3. Use predicted fragmentation and retention time for rescoring; control FDR within the SAAP class at 1%.
4. Exclude six-frame translation matches, mutation-compatible products, immunoglobulins, trypsin and problematic cleavage-site substitutions.

Confidence is not uniform

8,746 unique BP–SAAP pairs from 1,767 genes; only 1,955 have substitution-localization probability > 0.9.

Fig. 1; Methods: Candidate SAAP and PTM identification, Mass shift localization; Supplementary Tables 2 and 8.

Independent checks strengthen peptide identification

Spectral agreement

Median observed–predicted spectral angle: 0.845 for SAAPs versus 0.848 for BPs. Identification quality does not fall as RAAS rises.

Orthogonal contexts

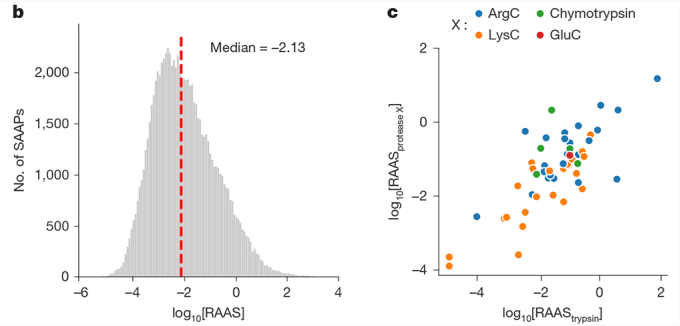
Multiple protease digests recover the same substitution sites. IP–MS identifies bait-derived SAAPs in 18 of 25 selected pull-downs.

Peptide-centric validation: PepQuery confidently matches about 78% of SAAPs, similar to BP controls; reversed SAAP controls match about 1%.

These checks support the assigned sequences, but do not prove every weakly localized mass shift is a single substitution.

Fig. 1h; Extended Data Fig. 2j,k; Supplementary Note 2 and Figs. 5–6. Spectral-angle scale: higher is better.

Most substitutions are minor; a substantial tail is abundant



About 10% have RAAS > 1.

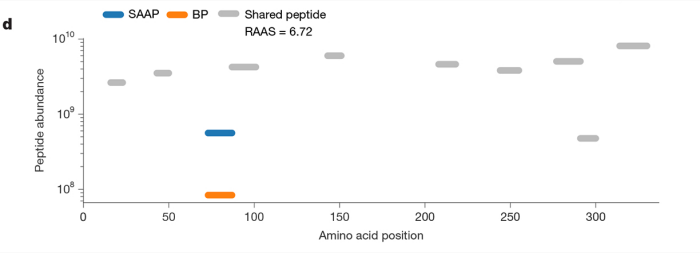
The authors infer a dominant substituted product for 360 proteins.

Across protease digests, ratio estimates agree: $r = 0.77$.

The plot reports $\log_{10}$ ratios, not fractions.

Fig. 2b,c (cropped, original axes retained). Left: abundance distribution; right: alternative proteases versus trypsin.

Do the other peptides agree with a dominant substituted product?



FBP1 example: RAAS = 6.72.

Shared peptides lie closer in abundance to the SAAP than to the BP.

The same pattern appears globally as RAAS increases.

Predicted ionization efficiency and detectability are similar for BP–SAAP pairs. These are supporting controls; response-factor differences and uncertain full proteoform composition remain caveats.

Fig. 2d (cropped); Fig. 2e; Extended Data Fig. 2d–i,n; Supplementary Note 3. Intensities are MS measurements.

Why abundance cannot be read as the decoding error rate

For encoded (P_1) and substituted (P_2) products, let S_i be synthesis rates and α_i first-order degradation rates:

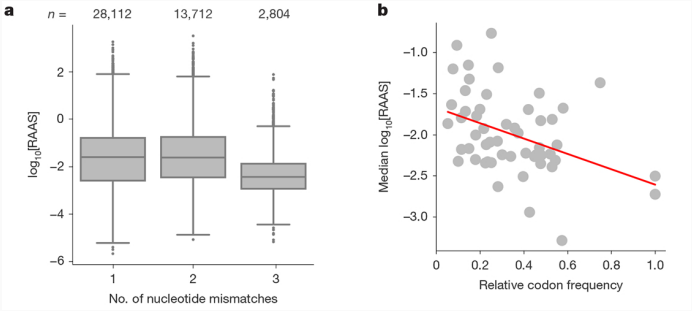
$$\frac{dP_i}{dt} = S_i - \alpha_i P_i \quad \Longrightarrow \quad \frac{P_2}{P_1} = \frac{S_2}{S_1}\frac{\alpha_1}{\alpha_2} \quad \text{at steady state.}$$

Illustrative example
A product synthesized at 1% of the encoded rate but removed ten times more slowly reaches a 10% abundance ratio.

A high ratio can reflect faster alternate synthesis, slower removal, or both. Stability differences of tenfold alone cannot turn a 1% synthesis ratio into RAAS > 1.

Fig. 2a; Supplementary Notes 4 and 6. Derivation assumes steady state, first-order turnover and comparable peptide response.

Codon-level associations support a translation contribution



More minimum codon–anticodon mismatches accompany lower RAAS.

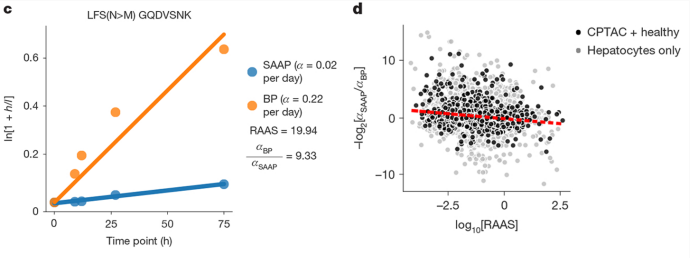
Rare codons have higher median RAAS: $r = -0.402$, 53 codons.

These are associations, not direct measurements of tRNA mispairing.

Uracil modification sites also overlap substitution codons (180 overlaps; $P < 10^{-10}$), and modification fractions correlate with RAAS. The modification data come from separate cell-line datasets.

Fig. 3a,b (cropped); Methods: Overlap with RNA modification sites; Supplementary Fig. 8 and Note 6.

Metabolic labelling links accumulation to slower turnover



Pulse with heavy amino acids; fit the heavy/light ratio over time:

$$\ln(1 + h/l) = \alpha t.$$

Some SAAPs turn over up to ten times more slowly than BPs.

Global relationship is significant but modest: $r = -0.16$ in hepatocytes.

Shared-peptide turnover provides a consistency check; B-cell and NK-cell datasets show similar inverse trends. The turnover datasets lack matched transcriptomes.

Fig. 3c,d (cropped); Fig. 3e,f; Extended Data Fig. 3m,n; Methods: SAAP degradation rates.

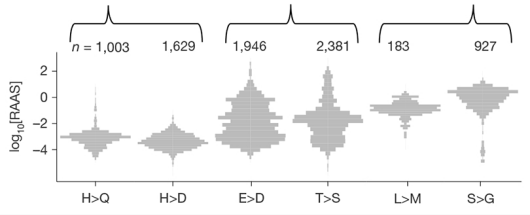
An important qualification: interventions are less decisive

- **Proteasome inhibition:** expected increases for less stable SAAPs were small and not statistically significant in the limited observed set.
- **Cycloheximide chase:** 86 SAAP–BP pairs were observed; one pair turned over rapidly enough for a clear comparison. Most SAAPs changed little during the eight-hour window.
- These results are compatible with stable products, but provide limited causal resolution for the global relationship between RAAS and stability.

How strongly should we interpret the mechanism?
Direct turnover estimates support stability differences. The relative contribution of synthesis versus degradation remains unresolved at most sites.

Supplementary Note 4 and Fig. 11; Methods: Degradation and translation inhibition analysis. Final sentence is journal-club interpretation.

The ratios depend strongly on the substitution and tissue



- Median RAAS spans 10^{-4} to above 1 across substitution types.
- Polar amino acids tend to yield higher ratios; some high-ratio types have few observations.
- Substitution type dominates variation, followed by tissue; G→S is enriched in pancreas in cancer and healthy datasets.

Fig. 4a (cropped; example distributions and sample counts); Fig. 4f; Extended Data Fig. 5h,i.

Biological context: suggestive associations rather than function tests

Cancer and protein organization

- Overall cancer/normal distributions are similar, but particular substitutions differ in matched comparisons.
- Lamin A is an example with elevated ratios in three tumour types.
- High ratios associate with disordered, less conserved regions and certain functional domains.

Population variation and species

- High-RAAS codons are less constrained in gnomAD; mutation-compatible SAAPs are excluded from the main set.
- Mouse analysis finds 397 unique SAAPs; 55 sequences are identical to human SAAPs.
- Shared human/mouse ratios correlate: $r = 0.55$.

Fig. 5a,f,h–k; Extended Data Figs. 6 and 9–10. Conservation and disease association do not establish adaptive function.

What limits the interpretation?

1. **Chemical ambiguity.** Incomplete fragment coverage leaves alternative modifications plausible for some assignments. Prioritize the 1,955 localized pairs; 557 have RAAS > 0.1.
2. **Detection selection.** Low-ratio sites on low-abundance proteins are often missing; observed RAAS distributions are not unbiased proteome-wide distributions.
3. **Peptides versus intact proteins.** Digestion, ionization and shared-peptide assignment affect inference; about 20% of SAAPs contain missed cleavages and deserve extra scrutiny.
4. **Mechanism and function.** No broad perturbation establishes programmed recoding or phenotypic effects. Tissue mixtures, synthesis and degradation can all influence ratios.

Discussion; Supplementary Notes 3, 6–8. Final tissue-mixture point is a journal-club interpretation.

What experiment would move the story forward?

1. Select fully cleaved, confidently localized high-RAAS pairs. Confirm both peptides with isotope-labelled standards and targeted MS; measure absolute response factors.
2. In one matched cell system, measure DNA, RNA, nascent synthesis and turnover. Perturb the implicated codon or RNA modification while preserving protein sequence where possible.
3. Reconstitute encoded and substituted proteins, quantify activity and lifetime, and test whether changing their ratio alters a cellular phenotype.

Take-home

The convincing observation is abundant, stable sequence-divergent peptide products. Programmed recoding and biological function are the next hypotheses to test.

Proposed experiments and interpretation, informed by the paper's Discussion and Supplementary Notes 6–8.

Sources and practical starting points

Version of record

Tsour S., Machné R., Leduc A. et al. *Nature* 656, 506–515 (2026). Published 24 June 2026.

`https://doi.org/10.1038/s41586-026-10678-2`

Full accepted manuscript and methods

`https://pmc.ncbi.nlm.nih.gov/articles/PMC13384240/`

Supplementary Information

Supplementary Notes 1–8 and Figs. 1–11 from the publisher's supplement.

Code, tables and data navigation

`https://decode.slavovlab.net` `https://github.com/SlavovLab/decode`

For follow-up, begin with Supplementary Table 8 (high localization confidence), rather than treating every candidate as equally secure.